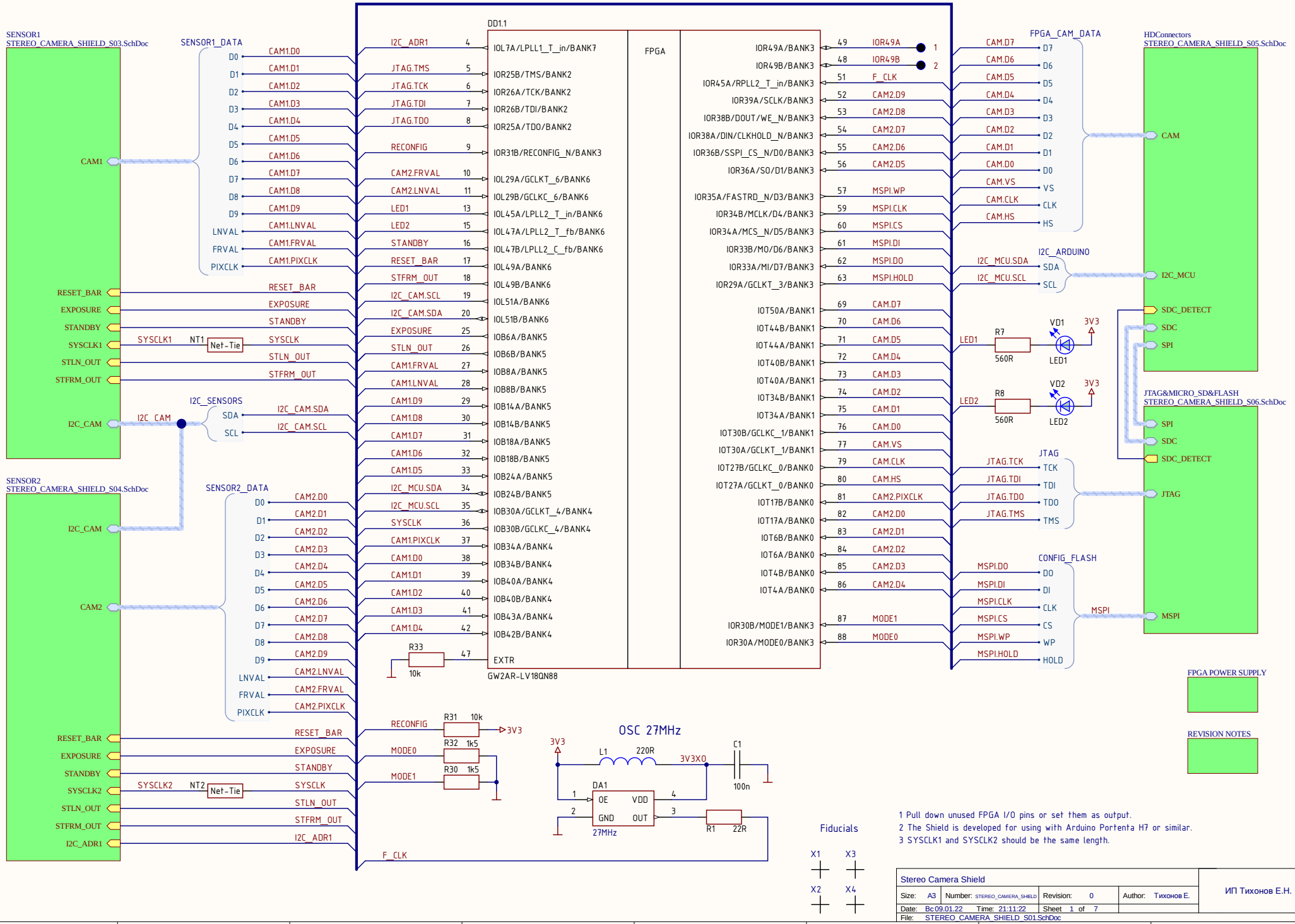
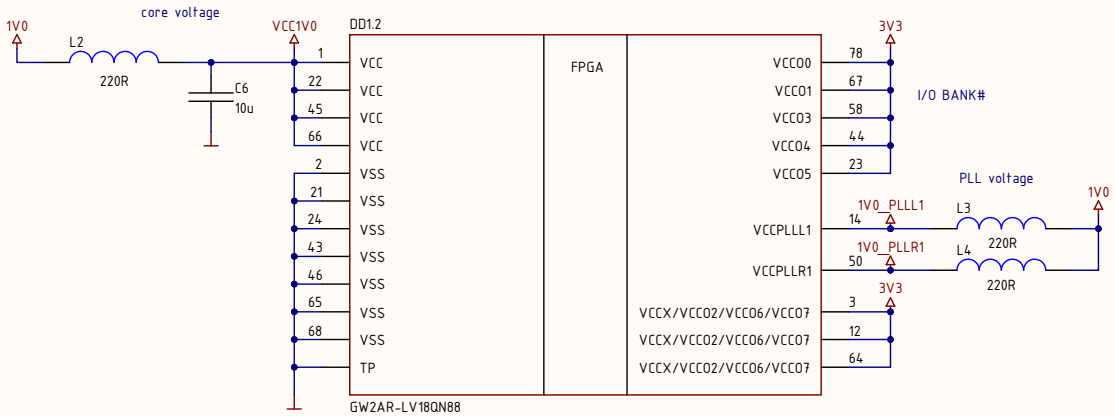
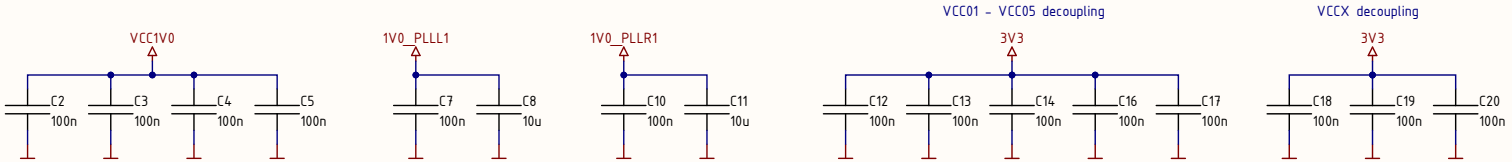
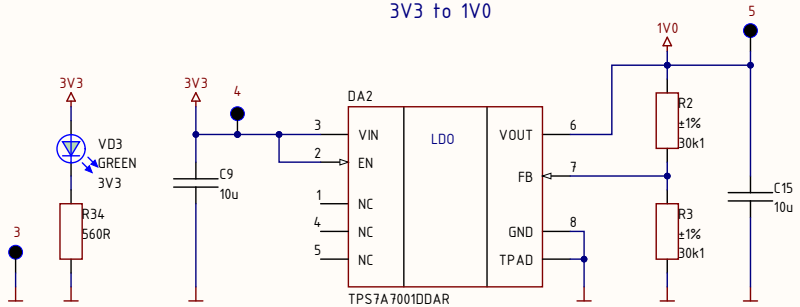

SENSOR1
STEREO_CAMERA_SHIELD_S03.SchDoc
SENSOR1_DATA
CAM1
RESET_BAR
EXPOSURE
STANDBY
SYSCLK1
STLN_OUT
STFRM_OUT
I2C_CAM
I2C_SENSORS
SENSOR2
STEREO_CAMERA_SHIELD_S04.SchDoc
SENSOR2_DATA
CAM2
SYSCLK2
I2C_ADR1
Net-Tie
DD1.1
FPGA
GW2AR-LV18QN88
EXTR
R33
10k
RECONFIG
MODE0
MODE1
R31 10k
R32 1k5
R30 1k5
3V3
OSC 27MHz
DA1
OE
VDD
GND
OUT
27MHz
L1 220R
3V3XO
C1
100n
R1 22R
F_CLK
FPGA_CAM_DATA
HDConnectors
STEREO_CAMERA_SHIELD_S05.SchDoc
CAM
I2C_ARDUINO
I2C_MCU
SDC_DETECT
SDC
SPI
VD1
VD2
LED1
LED2
R7
R8
560R
JTAG&MICRO_SD&FLASH
STEREO_CAMERA_SHIELD_S06.SchDoc
JTAG
CONFIG_FLASH
MSPI
FPGA POWER SUPPLY
REVISION NOTES
Fiducials
X1
X2
X3
X4
1 Pull down unused FPGA I/O pins or set them as output.
2 The Shield is developed for using with Arduino Portenta H7 or similar.
3 SYSCLK1 and SYSCLK2 should be the same length.
Stereo Camera Shield
Size: A3
Number: STEREO_CAMERA_SHIELD
Revision: 0
Author: Тихонов Е.
Date: Вс 09.01.22
Time: 21:11:22
Sheet 1 of 7
File: STEREO_CAMERA_SHIELD_S01.SchDoc
ИП Тихонов Е.Н.

FPGA POWER SUPPLY

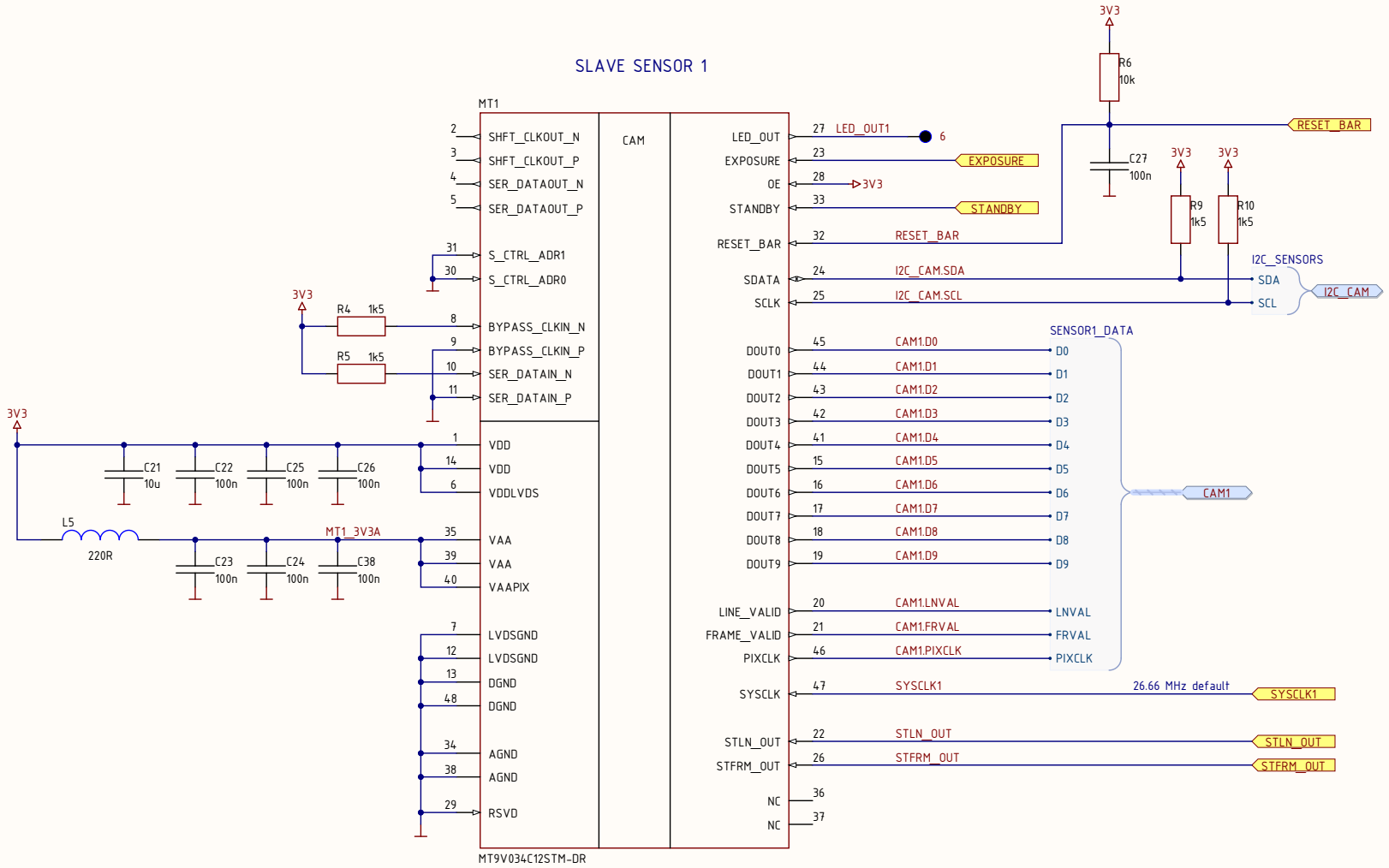




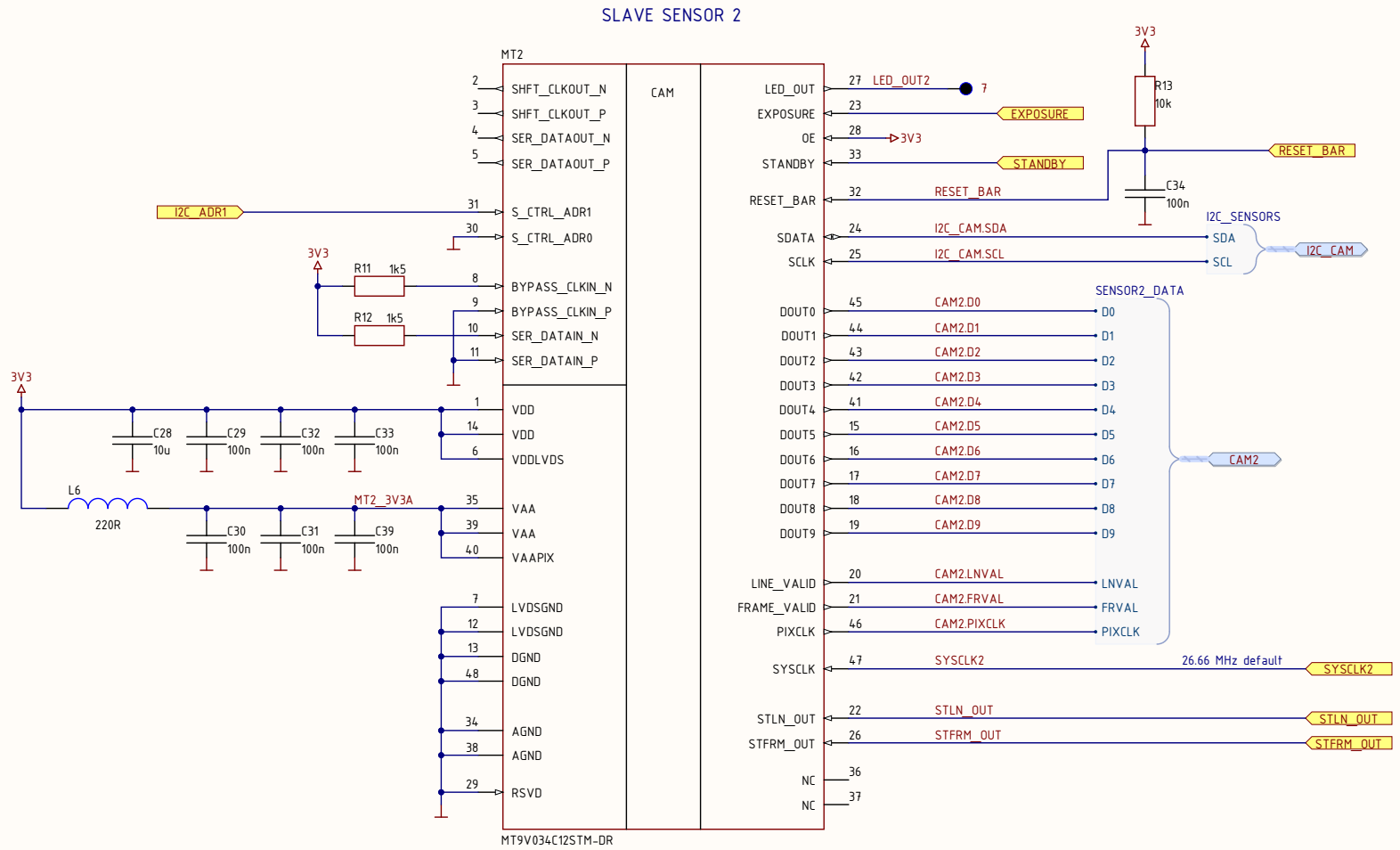
3V3 to 1V0



Stereo Camera Shield				ИП Тихонов Е.Н.
Size: A3	Number: STEREO_CAMERA_SHIELD	Revision: 0	Author: Тихонов Е.	
Date: Вс 09.01.22	Time: 21:11:23	Sheet 2 of 7		
File: STEREO_CAMERA_SHIELD_S02.SchDoc				



Stereo Camera Shield				ИП Тихонов Е.Н.
Size: A3	Number: STEREO_CAMERA_SHIELD	Revision: 0	Author: Тихонов Е.	
Date: Вс 09.01.22	Time: 21:11:23	Sheet 3 of 7		
File: STEREO_CAMERA_SHIELD_S03.SchDoc				


SLAVE SENSOR 2
MT2
CAM
MT9V034C12STM-DR
SHFT_CLKOUT_N
SHFT_CLKOUT_P
SER_DATAOUT_N
SER_DATAOUT_P
S_CTRL_ADR1
S_CTRL_ADR0
BYPASS_CLKIN_N
BYPASS_CLKIN_P
SER_DATAIN_N
SER_DATAIN_P
VDD
VDD
VDDLVDS
VAA
VAA
VAAPIX
LVDSGND
LVDSGND
DGND
DGND
AGND
AGND
RSVD
LED_OUT
EXPOSURE
OE
STANDBY
RESET_BAR
SDATA
SCLK
DOUT0
DOUT1
DOUT2
DOUT3
DOUT4
DOUT5
DOUT6
DOUT7
DOUT8
DOUT9
LINE_VALID
FRAME_VALID
PIXCLK
SYSCLK
STLN_OUT
STFRM_OUT
NC
NC
I2C_ADR1
3V3
R11 1k5
R12 1k5
C28 10u
C29 100n
C32 100n
C33 100n
L6
220R
MT2_3V3A
C30 100n
C31 100n
C39 100n
R13 10k
C34 100n
LED_OUT2
7
EXPOSURE
3V3
STANDBY
RESET_BAR
I2C_SENSORS
SDA
SCL
I2C_CAM
I2C_CAM.SDA
I2C_CAM.SCL
SENSOR2_DATA
CAM2.D0
CAM2.D1
CAM2.D2
CAM2.D3
CAM2.D4
CAM2.D5
CAM2.D6
CAM2.D7
CAM2.D8
CAM2.D9
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
CAM2
CAM2.LNVAL
CAM2.FRVAL
CAM2.PIXCLK
LNVAL
FRVAL
PIXCLK
SYSCLK2
26.66 MHz default
SYSCLK2
STLN_OUT
STFRM_OUT
LENS2
M12 Lense holder
Stereo Camera Shield
Size: A3
Number: STEREO_CAMERA_SHIELD
Revision: 0
Author: Тихонов Е.
Date: Вс 09.01.22
Time: 21:11:23
Sheet 4 of 7
File: STEREO_CAMERA_SHIELD_S04.SchDoc
ИП Тихонов Е.Н.

J1 LEFT ARDUINO [1)] CONNECTOR

XS1 — LEFT

Цепь			
1	ETH A+	NC	2
3	ETH A-	NC	4
5	ETH B+	NC	6
7	ETH B-	NC	8
9	NC	DSI D1+	10
11	NC	DSI D1-	12
13	NC	DSI D0+	14
15	NC	DSI D0-	16
17	ETH L1	DSI CK+	18
19	ETH L2	DSI CK-	20
21	VIN	GND	22
23	USB1 VBUS	NC	24
25	USB1 D+	USB0 D+	26
27	USB1 D-	USB0 D-	28
29	USB1 ID	NC	30
31	GND	VIN	32
33	UART1 TX	UART0 TX	34
35	UART1 RX	UART0 RX	36
37	UART1 RTS	UART0 RTS	38
39	UART1 CTS	UART0 CTS	40
41	VIN	GND	42
43	I2C1 SDA	I2C0 SDA	44
45	I2C1 SCL	I2C0 SCL	46
47	GND	VIN	48
49	CAN1 TX	NC	50
51	CAN1 RX	NC	52
53	VSYS	GND	54
55	SDC CLK	I2S MCK	56
57	SDC CMD	I2S WS	58
59	SDC D0	I2S DI	60
61	SDC D1	I2S DO	62
63	SDC D2	VSYS	64
65	SDC D3	DMIC CK	66
67	NC (SDC CD)	DMIC D0	68
69	NC	NC	70
71	NC	VCC/3V3	72
73	NRST	NC	74
75	SWDIO	NC	76
77	SWCK	NC	78
79	SWO	NC	80

DF40HC(3.5)-80DS-0.4V(51)

I2C_MCU.SDA → I2C_ARDUINO.SDA
I2C_MCU.SCL → I2C_ARDUINO.SCL
I2C_ARDUINO → I2C_MCU

SD_CARD: CLK ← SDC.CLK (55), CMD ← SDC.CMD (57), D0 ← SDC.D0 (59), D1 ← SDC.D1 (61), D2 ← SDC.D2 (63), D3 ← SDC.D3 (65), SDC_DETECT (67)
SDC

VCC/3V3 (72) → 3V3

J2 RIGHT ARDUINO [1)] CONNECTOR

XS2 — RIGHT

Цепь			
1	BOOT	CAM D7	2
3	BOOT SEL	CAM D6	4
5	PWRON	CAM D5	6
7	COINCELL	CAM D4	8
9	NC	CAM D3	10
11	NC	CAM D2	12
13	NC	CAM D1	14
15	NC	CAM D0	16
17	NC	CAM VS	18
19	NC	CAM CLK	20
21	NC	CAM HS	22
23	VCC/3V3	GND	24
25	UART3 TX	UART2 TX	26
27	UART3 RX	UART2 RX	28
29	NC	NC	30
31	NC	NC	32
33	GND	VCC	34
35	NC	SPI1 CS	36
37	NC	SPI1 CK	38
39	NC	SPI1 MISO	40
41	NC	SPI1 MOSI	42
43	VCC/3V3	GND	44
45	I2C2 SDA	GPIO 0	46
47	I2C2 SCL	GPIO 1	48
49	SAI CK	GPIO 2	50
51	SAI FS	GPIO 3	52
53	SAI D0	GPIO 4	54
55	NC	GPIO 5	56
57	GND	GPIO 6	58
59	PWM 1	PWM 6	60
61	PWM 2	PWM 7	62
63	PWM 3	PWM 8	64
65	PWM 4	PWM 9	66
67	PWM 5	PWM 10	68
69	VCC/3V3	GND	70
71	ADC VREF+	ADC VREF-	72
73	ADC A0	ADC A4	74
75	ADC A1	ADC A5	76
77	ADC A2	ADC A6	78
79	ADC A3	ADC A7	80

DF40HC(3.5)-80DS-0.4V(51)

3V3 → VCC/3V3 (23), VCC/3V3 (43), VCC/3V3 (69)
3V3 ← VCC (34)

FPGA_CAM_DATA: CAM.D7 (2) … CAM.HS (22) → CAM: D7, D6, D5, D4, D3, D2, D1, D0, VS, CLK, HS

ARDUINO_SPI: SPI.CS (36), SPI.CK (38), SPI.MISO (40), SPI.MOSI (42) → SPI: CS, CK, MISO, MOSI

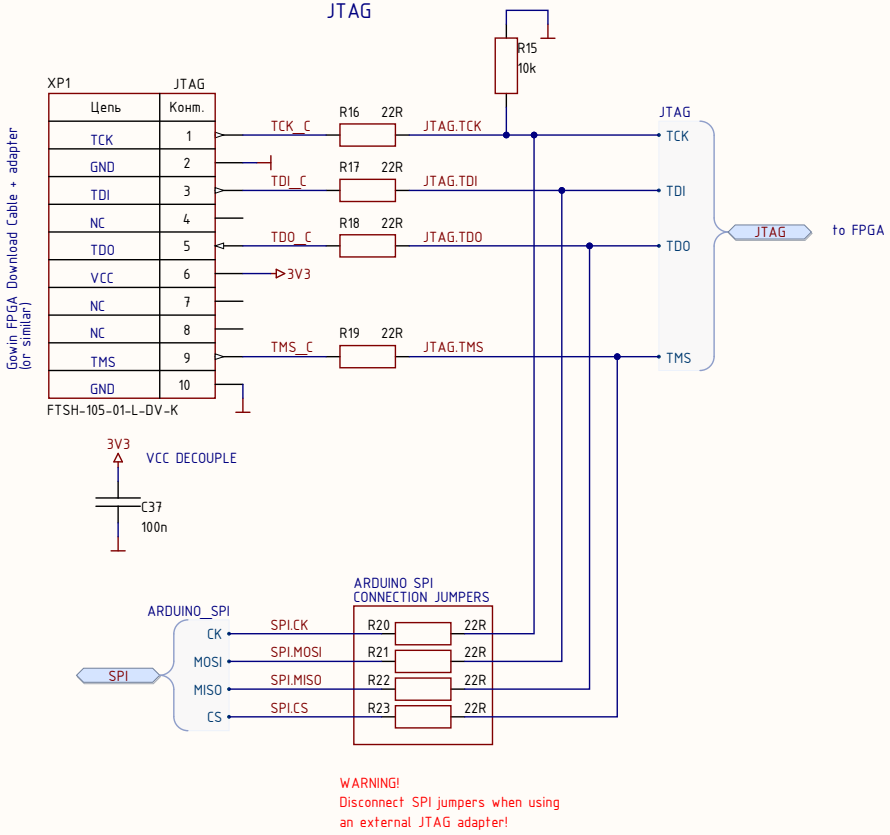
SDC_DETECT (46) → SDC_DETECT

NOTES:

1. Arduino Portenta H7

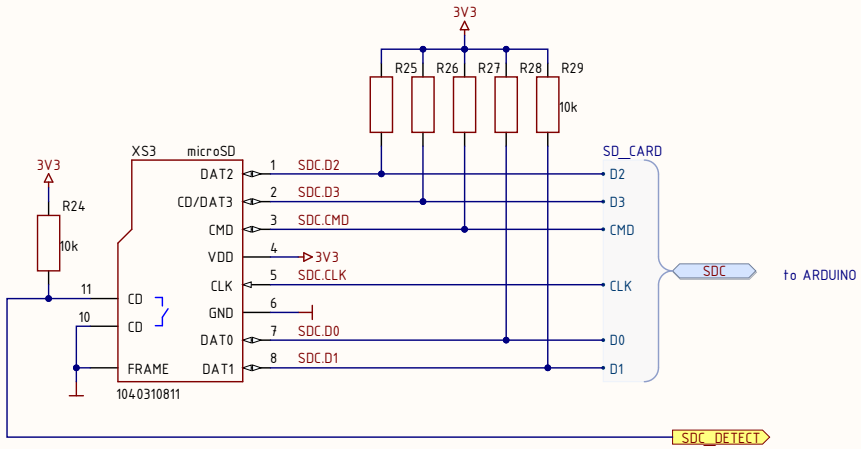
Stereo Camera Shield				ИП Тихонов Е.Н.
Size: A3	Number: STEREO_CAMERA_SHIELD	Revision: 0	Author: Тихонов Е.	
Date: Вс 09.01.22	Time: 21:11:23	Sheet 5 of 7		
File: STEREO_CAMERA_SHIELD_S05.SchDoc				

JTAG

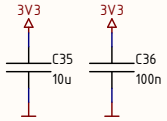


WARNING!
Disconnect SPI jumpers when using an external JTAG adapter!

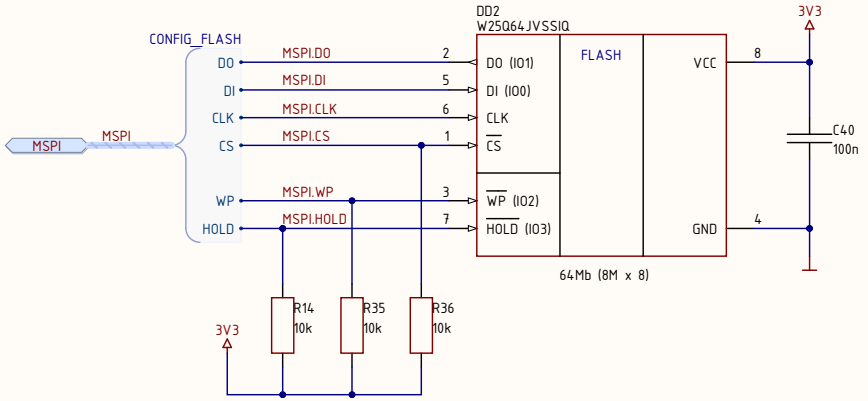
MICRO SD CARD



SD CARD DECOUPLE



FPGA CONFIG FLASH



Stereo Camera Shield				ИП Тихонов Е.Н.
Size: A3	Number: STEREO_CAMERA_SHIELD	Revision: 0	Author: Тихонов Е.	
Date: Вс 09.01.22	Time: 21:11:23	Sheet 6 of 7		
File: STEREO_CAMERA_SHIELD_S06.SchDoc				

REVISION NOTES

PCBA	PCB	Description	Date	Author
0	0	Прототип	08.01.2022	Tikhonov

Stereo Camera Shield				ИП Тихонов Е.Н.
Size: A3	Number: STEREO_CAMERA_SHIELD	Revision: 0	Author: Тихонов Е.	
Date: Вс 09.01.22	Time: 21:11:23	Sheet 7 of 7		
File: STEREO_CAMERA_SHIELD_REV.SchDoc				